Article

Project Safe Wireless: From EM-Calibrated RSS Tomography to Traceable Propagation and a Real CSI Presence Pilot

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Abstract

Project Safe Wireless tests whether EM-based models of RSS noise improve radio tomographic localization. This revision preserves the original synthetic RTI study, its disjoint data phases, K-NN, room-center baseline, and equivalent-regularization control. It extends the same project with Sionna propagation, artificial CSI controls, and physical ESP32 acquisition for a related presence task. The original 30-query pilot did not outperform the center baseline, and Fisher-scaled RTI matched its regularization control. A binary random forest with 200 trees was evaluated on 165 valid two-second CSI windows from three rounds in one environment. Leaving one round out gave 70.13% positive predictive value, 61.90% still-person sensitivity, 71.79% walking sensitivity, and 27.38% false positives in empty conditions. An exploratory comparison on identical windows and folds gave balanced accuracy of 69.64% for the forest, 61.93% for logistic regression, and 50.00% for a training-majority control. It was added after inspecting the forest and is not independent confirmation. The contribution is an expanded, traceable workflow rather than demonstrated superiority over RSS/RTI, other repositories, or rescue sensing. Round instability and absence of new-room evaluation preclude a reliable-deployment claim.

Keywords: Project Safe Wireless; expectation-maximization; radio tomographic imaging; RSS; channel state information; ESP32; presence detection; reproducibility

1. Introduction

Device-free radio localization estimates an object's position without asking it to carry a transmitter. In RTI, changes in RSS across radio links are modelled as weighted contributions from cells in a spatial grid. The forward model is

$$\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{n}, \tag{1}$$

where $\mathbf{y}$ contains the link measurements, $\mathbf{W}$ maps grid cells to links, $\mathbf{x}$ is the unknown attenuation field, and $\mathbf{n}$ is measurement noise. The number of links is often smaller than the number of cells, so the reconstruction is ill posed and needs regularization [1].

The long-term motivation is to investigate radio-based sensing that could support the search for people under rubble. Prior rescue research has studied breathing-related motion with radar [2]. The present work is an initial computational step: it localizes an idealized attenuation target and leaves rubble and material properties outside the model. The simulation does not establish whether a person is present or whether radio sensing would work in a collapsed structure.

RTI estimates position from indirect RSS measurements, so its result depends on both the spatial model and the assumed noise. Expectation-Maximization (EM) can fit a two-component Gaussian mixture to RSS residuals, but a better statistical fit does not guarantee a more accurate position. The practical question is whether that improvement in noise modeling carries through to localization. Node geometry, regularization, calibration size, and target strength also affect the answer.

The objective of this study is to determine whether better statistical modelling of RSS noise leads to better position estimates, and by how much. From an AI standpoint, this is a question about unsupervised probabilistic modelling: whether a mixture model learned from unlabeled calibration residuals improves a downstream prediction.

This project tests that question under controlled synthetic conditions. EM estimates the mixture parameters and the latent component probabilities from calibration residuals; a separate validation block measures noise-model fit, and held-out test queries measure localization. The research question is: across the nine predefined synthetic scenarios, does a two-Gaussian mixture fitted by EM achieve higher held-out log density than a single Gaussian, and how does its Fisher-derived scaling affect localization error relative to standard Tikhonov RTI? What does the equivalent-regularization control reveal?

The study reports EM convergence, held-out noise density, and localization error as distinct outcomes. It uses independent data partitions and compares against Tikhonov RTI, K-NN, and a center prediction. Its present contribution is an integrated, reproducible, scenario-based comparison of noise fitting and localization that can serve as a starting point for later rescue-oriented research. The new element is the controlled test that separates noise-model fit from localization error and includes a regularization-matched control, so that any gain can be attributed to the noise model rather than to a change in penalty strength. It combines established ideas and software; any future claim of novelty for rescue sensing would require comparison and experimental evidence [1–4].

This V2 revision follows the course's Midterm Project Proposal requirements and continues the V1 research paper, rather than replacing it with a separate software report [15]. The original AI question about learned noise distributions and downstream localization is retained. The new applied question is whether physical CSI processed by a simple learned model can distinguish absence and presence, including a still person, on complete held-out acquisition rounds. Presence classification and spatial localization remain different outcomes. The folder `safe-wirreles` identifies the implementation extension of Project Safe Wireless.

2. Related Work

Dempster, Laird, and Rubin described EM as a general maximum-likelihood method for models with incomplete data, including finite mixtures [5]. Wu later analysed when EM sequences converge [6]. In practice, reaching the iteration limit or returning a stable parameter vector does not establish that the fitted model suits the data or improves a downstream prediction.

Wilson and Patwari introduced a linear RSS-based RTI model and used Tikhonov regularization to reconstruct attenuation images [1]. They also fitted a two-component Gaussian mixture to measured RSS residuals with EM. The base simulation uses their reported values, $p = 0.548$, $\sigma_1 = 0.971$ dB, and $\sigma_2 = 3.003$ dB, as fixed generator settings. This project does not measure those parameters. Their experiment used IEEE 802.15.4 sensor nodes, so its measurements are not a Wi-Fi dataset.

These RTI and noise parameters [1], the general EM formulation [5], and the convergence analysis [6] are the three main scientific foundations of the present implementation. Other studies below inform its comparisons and future direction.

Later RTI studies combined mixture models with reconstruction methods. Gao et al. used a Gaussian mixture model and matrix-transform restoration, and evaluated it with simulated and experimental targets [3]. Huang and Yang estimated mixture-noise parameters and the attenuation image together through sparse Bayesian learning (SBL) [4]. Zhao and Patwari addressed a related problem in variance-based device-free localization: they either projected the measurements away from a noise subspace learned from calibration data or weighted them with an estimated link-noise covariance, and reported lower localization RMSE than the standard Tikhonov-based estimator in two measured experiments [7]. In this implementation, the Fisher term is one scalar that changes the Tikhonov regularization. It does not assign different noise weights to individual links, as a link-adaptive estimator would.

Kaltiokallio et al. used EM with Gaussian smoothing to learn RSS model parameters for device-free localization and tracking, then evaluated the method on measured data from two environments [8]. Their experiments track moving targets; this project uses static synthetic queries on one grid. RADAR provides an earlier example of RF localization from signal strengths and known locations [9]. The K-NN method here is a weighted fingerprint baseline with synthetic data, rather than a reproduction of RADAR.

For the intended rescue context, Takeuchi et al. studied a radar system that used breathing-related motion to locate survivors under rubble [2]. Its sensing modality and experimental setting differ from the RSS-based synthetic localization studied here. The software also draws on IsukaW's Wi-Fi sensing repository [10]: the current project reuses its WiFiMeasurement data contract, while the synthetic generator, EM calibration, and RTI evaluation form a separate integration. The repository is cited as software provenance, not as experimental evidence for rescue performance. Table 1 summarizes the related studies.

The extension distinguishes activity from presence. CARM connects CSI dynamics to activity recognition and reports an average accuracy above 96% in its experiments [11]; this is not stationary-presence accuracy. Liu et al. combine CSI magnitude and phase CNN branches with temporal post-processing [12]. Their author manuscript reports collection on 24 days in three environments, but states that complete stillness is likely classified as no motion before post-processing. Neither validates our ESP32 hardware or still-person protocol, and their scores are not transferred to this study.

IsukaW remains software provenance rather than a performance benchmark. The remote audit resolved revision `91aee04ea938274d014a3981d409f83267ea15c0`; an earlier local copy was `065c12be3b0fb34d458d74d36d3534746e285f84`. Its Wi-Fi path uses RSSI and heuristic radar coordinates; CSI is declared but no connected collector was identified. Camera-assisted fusion supplies an additional modality. RTI/EM belong to Project Safe Wireless, not that reference. No measured performance ranking against it is claimed.

Table 1. Summary of related work.

Reference	Main Idea / Problem	Model / Algorithm	Dataset	Metric(s)	Main Result(s)	Limitation(s)
Wilson & Patwari [1]	Infer attenuation images from RSS changes across links.	Linear RTI weights, Tikhonov reconstruction, and EM fit of two-Gaussian RSS noise.	Deployed 28-node sensor network over 441 ft^2.	Image MSE and geometry-based MSE bounds.	Demonstrates RTI reconstruction with measured RSS; accuracy depends on geometry and parameter choice.	IEEE 802.15.4 deployment; it does not isolate EM's effect on downstream position error.
Gao et al. [3]	Restore blurred RTI images to recover target shape.	Gaussian-mixture degradation model, matrix transform, constrained least-squares filter.	Simulated target shapes and an indoor RTI experiment.	Image MSE, edge-preservation index, and SSIM.	Reports improved target detail and image-quality measures over shadowing-based RTI.	Focuses on image restoration, not noise calibration or position-error gains.
Huang & Yang [4]	Reduce mislocalization under complex, non-Gaussian RTI noise.	Sparse Bayesian learning jointly estimates a Gaussian-mixture noise model and attenuation image.	20-node, 6 m × 4 m setup; predefined single- and multi-target positions.	Mean and maximum localization error; multi-target set-distance.	SBL-NA outperformed Tikhonov, BCS, HBCS, and Laplace baselines in reported localization errors; best tested mixture used $K = 3$.	Joint SBL reconstruction is more complex and differs from this study's two-Gaussian calibration control.
Zhao & Patwari [7]	Reduce the effect of intrinsic motion noise (e.g., wind) in variance-based device-free localization.	Variance-based RTI with a subspace projection (SubVRT) or covariance-weighted least squares (LSVRT).	34 TelosB nodes and 1122 links around a house; two measured experiments (calm and windy).	Localization and tracking RMSE.	Windy case: RMSE 1.26 m (VRTI) versus 0.74 m (SubVRT) and 0.69 m (LSVRT); small gains in the calm case.	Detects motion rather than static attenuation; LSVRT assumes Gaussian noise; needs calibration data.
Kaltiokallio et al. [8]	Learn RSS parameters online while tracking device-free targets.	Gaussian smoothing with EM and Bayesian filtering with measurement selection.	18 experiments in a 75 m^2 open area and an 82 m^2 apartment.	Tracking RMSE and computation time.	Reports 17 cm and 37 cm tracking accuracy; 42% lower RMSE than ARTI and EM up to 18× faster than nonlinear optimization.	Dynamic measured-RSS tracking, not static synthetic RTI localization.
Bahl & Padmanabhan [9]	Locate building occupants from WLAN signal strengths.	Empirical nearest-neighbor fingerprints and a propagation-model baseline.	Indoor WLAN deployment with three base stations and known-location samples.	Percentile and median position error.	Reports a 2–3 m median for empirical fingerprints, versus 4.3 m for propagation modeling.	Site-specific calibration; older WLAN setup; no tomography or EM noise estimation.
Takeuchi et al. [2]	Detect breathing motion to locate survivors under rubble.	Pulse-Doppler rescue radar with a two-dimensional array and 3D location estimation.	Experimental rescue-radar evaluation; the article abstract does not give a sample count.	Experimental feasibility of 3D survivor localization; not directly comparable to RSS RMSE.	The authors report experimental assessment of the array-based approach.	Different sensing modality and target signal; motivation only, not an RTI benchmark.
Wang et al. [11]	Relate CSI dynamics to activities.	CSI–speed and CSI–activity models (CARM).	Commercial Wi-Fi in multiple environments; counts not reconciled here.	Activity accuracy.	Reports average accuracy above 96%.	Activity recognition does not establish still presence or ESP32 performance.
Liu et al. [12]	Presence from ambient Wi-Fi.	Magnitude/phase CNN and temporal post-processing.	Atheros CSI; 24 days, three environments, eight months.	Motion/presence tests; false-positive duration.	Four false-positive seconds in about 46.5 h of human-free test.	Completely still persons may be negative for the motion model; hardware/protocol differ.

The reviewed studies show that link-wise noise handling can help on measured data [4,7], but they largely change the estimator and the noise model together, and the EM fit of Wilson and Patwari [1] was not evaluated against position error. What remains open is whether an improvement in noise-model fit alone carries through to localization, and under which noise conditions (correlation, quantization, missing links) it does. This study uses standard Tikhonov RTI [1] as the benchmark method, with weighted K-NN [9] and a room-center prediction as additional baselines. Covariance-weighted RTI [7] is the natural stronger benchmark for the final version.

3. Materials and Methods

3.1. Study Design and Data Generation

The V1 fundamental investigation of an EM-fitted noise model is retained. V2 additionally evaluates an applied presence-classification model against simpler controls on measured CSI; the tasks and outcomes are reported separately.

The V1 layer is a fundamental-research study: it investigates whether an EM-fitted noise model improves a standard RTI algorithm, rather than solving a deployed application. It is an in silico study using simulated measurements. The generator places radio nodes around a rectangular area, builds the ellipse-based weight matrix $\mathbf{W}$, draws one target attenuation field $\mathbf{x}$, and generates link changes using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{n}$. It records RSS as the reference level minus the generated attenuation. Each CSV row identifies its link and data phase; missing measurements are marked explicitly, and the target position is saved separately as ground truth.

The base mixture uses Wilson and Patwari's published values [1]. The quick run has six nodes, 15 links, 36 grid cells, and 30 test queries. The larger base run uses 12 nodes, 66 links, 100 cells, and 120 test queries. Both configurations save separate reference, calibration, validation, fingerprint, development, and test data. The simulator can vary noise spread, quantization, temporal correlation, link loss, baseline drift, carrier frequency, and receiver margin. In this model, carrier frequency changes the illustrative free-space link budget and which weak links are censored; it does not change the spatial weight matrix or simulate signal penetration through materials.

The system workflow (Figure 1) connects synthetic data generation, EM noise calibration, held-out noise validation, localization, and evaluation. Known fingerprint positions are used by K-NN and development positions select hyperparameters; the true positions of test queries are reserved for final evaluation.

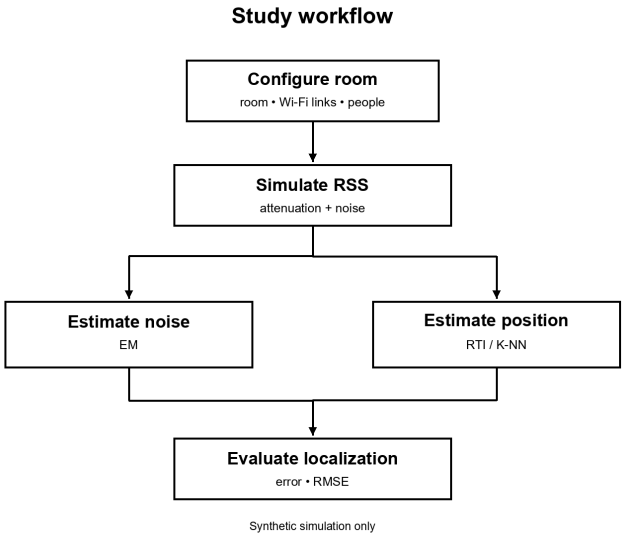


Figure 1. Study workflow. The simulated room, links, and people are configured; RSS is simulated with attenuation and noise; EM estimates the noise model while RTI or K-NN estimates position; both feed the evaluation of localization error and RMSE. Synthetic simulation only.

The V1 design is retained as the synthetic RTI layer. The new propagation layer uses Sionna RT 1.2.2, Mitsuba 3.8.0 and Dr.Jit 1.3.1 [13]. It calculates paths, CIR, CFR and radio maps; local checks include LoS/Friis and absolute delay. Two static channels are adapted to an array of shape $(1, 64, 10)$ with no class prediction. These ideal frequency bins are not chipset CSI. Observation noise does not create gait or respiration, and bodies/materials remain illustrative. A Windows `LIVE_RSSI` observation with system-reported dBm verifies intensity acquisition only. RF interference and quantitative low/medium noise profiles remain unvalidated.

A separate `SYNTHETIC (fake)` development source produced 240 designed windows in four classes and three arbitrary domains called rooms. They are neither measured rooms nor Sionna scenes. This source supplies numerical and grouped-evaluation software controls, not physical classification evidence.

Physical acquisition used ESP32-D0WD-V3 revision v3.1, ESP-IDF v6.1 and the official `esp-csi` router example at commit `8633d67152db2808f141cc1595970aa9cf406045`, with a documented local compatibility patch [14]. The first unlabeled check retained 4,985 valid packets in 30.032 s. Packet times were irregular. The four initial bytes marked invalid were preserved raw and excluded from analysis; no physical subcarrier frequencies were assigned to payload indices.

Figure 1 is the original student-designed Figma diagram documented in V1 and is retained unchanged. It represents the synthetic workflow. The course requires the system diagram to be designed by the student, not generated by AI [15]. The student must update it to cover the CSI extension before claiming it depicts the complete V2 system; a measured-results plot is not a substitute.

3.2. Inputs, Partitions, and Algorithms

The estimator receives calibration residuals after the reference estimate has been subtracted. The C++17 EM code estimates the mixture weight p and both standard deviations while holding the component means at zero. A separate validation block compares the fitted mixture's average held-out log density with that of a single Gaussian. This measures how well each distribution predicts residuals; it does not measure localization accuracy.

The generator uses separate random streams for reference, calibration, noise validation, fingerprints, development, and test data. Development data selects regularization α and K; test data is reserved for scoring. K-NN fingerprints cover the same grid as the test locations but have independent noise draws. The current experiment therefore tests localization within a known grid. It does not test unseen positions, moving trajectories, multiple targets, or transfer to a different environment.

The localization comparison includes five methods: (1) Tikhonov RTI, (2) RTI with a Fisher factor calculated from the EM mixture, (3) a Tikhonov control with the equivalent regularization, (4) inverse-distance weighted K-NN on synthetic fingerprints, and (5) a constant prediction at the room center. In this implementation, the Fisher factor only rescales the Tikhonov penalty. The matched control checks that algebra; the equality does not demonstrate link-specific noise handling.

The physical pilot acquired three rounds, each with a separate operational empty reference, then empty, walking, still and final-empty recordings. Conditions were confirmed by the student, without external verification of labels, positions or distances. A previously fixed packet format was used. Amplitudes were normalized to the median empty reference. Two-second windows were resampled to 50 Hz from actual times without extrapolation or gaps over 100 ms. Calibration windows were not scored as empty examples.

The physical classifier uses twelve summaries of log amplitude ratios, static profiles and temporal variation. Settings were fixed at 200 trees, maximum depth 5, minimum leaf size 3, balanced class weights, seed 0, one worker and threshold 0.5. Each of three folds fits two complete rounds and holds out the third. That round uses its own separate empty reference: this is evaluation after environmental calibration, not transfer without a reference. Forest votes are uncalibrated scores. The artificial four-class control is different: separate synthetic rooms fit the forest, calibrate a sigmoid and test.

3.3. Metrics and Planned Experiments

For noise fit, the study records EM convergence, estimated parameters, and mean held-out log density (natural log per residual; residuals are in dB). For localization, it reports mean Euclidean error, RMSE in meters, and coverage. Mean, median, and RMS position error are the metrics reported by the RTI and RF-localization studies reviewed above

[4,7–9], and the held-out log density measures how well a fitted distribution predicts unseen residuals. The saved tables identify missing predictions and any center fallback. The V1 synthetic pilot uses one seed. The final evaluation will repeat the base and stress scenarios with independent seeds, summarize variation across runs, and compare methods on the same test queries.

The next stage will first reproduce the nine-scenario pilot, then repeat it across seeds to measure variability and convergence. The larger base geometry will also be run with several seeds. Correlated-noise and low-margin cases will be reported as failures or insufficient calibration whenever the estimator cannot fit the data. Each run saves its configuration, code hashes, generated CSVs, predictions, and summary. The main pipeline uses Python, NumPy, Matplotlib, and the existing C++17 EM code. Optional pandas and SciPy are only needed for diagnostics adapted from the upstream repository.

The updated four-week schedule (Table 2) preserves the original research direction while adding physical validation. A fifth week remains optional. Proposed tasks are distinct from the preliminary results already observed.

Table 2. Planned work schedule.

Week	Work	Deliverable
1	Reproduce V1 and freeze RTI/CSI partitions, parameters and baseline comparison.	Versioned protocol, hashes and reproduction record.
2	Acquire still-person and non-human disturbance controls; record geometry; develop position-likelihood RTI separately.	New measurements and separately tested estimators.
3	Repeat RTI seeds and CSI evaluation across days/environments with frozen models and thresholds.	Paired comparisons, convergence records and failures.
4	Integrate analysis, student-updated system diagram, limitations and reproducibility archive.	Reviewed MDPI paper and corresponding repository.
5 (optional)	Examine a compatible public RSS dataset as an external check.	Separate exploratory validation, if data format permits.

The schedule starts after the preliminary stage completed on 30 October 2026. Weeks 1–4 correspond to 31 October–6 November, 7–13 November, 14–20 November and 21–27 November; the optional fifth week is 28 November–4 December. These are proposed research dates, not completed validation. The available resources are the identified ESP32, preserved firmware and Sionna environment, in addition to V1's Python/C++17 tools.

Presence metrics are PPV, recall, F1, balanced accuracy and empty-condition false-positive rate, with separate still/walking recall. Spatial error in meters and classification percentages answer different questions and are not pooled into a single improvement score. Applicable consent, privacy and institutional requirements must be checked for physical observations.

3.4. Baselines and Validity Checks

Tikhonov RTI is the main algorithmic baseline, following the foundational RTI formulation [1]. Its equivalent-regularization control checks whether the Fisher factor does anything beyond changing the penalty strength. Weighted K-NN provides a fingerprint comparison inspired by RF location systems [9], and the room-center prediction checks

whether the signal-based methods improve on a simple spatial prior. Calibration and validation data are disjoint. Hyperparameters are selected with development data and are fixed before test scoring.

The simulator also checks that CSV values survive a write-read cycle, link identities remain consistent, missing values stay marked, and each run records its parameter sources. Correlated noise violates the independence assumption of the base mixture model, and EM may fail to converge under that condition. When a low receiver margin leaves too few calibration residuals, the run stops before fitting. In either case, the summary reports the failure or insufficient calibration and omits a localization score when the pipeline cannot validly produce one.

For the new applied task, two simpler baselines were evaluated on exactly the same 165 accepted windows, twelve features, empty references and three held-out-round partitions. The majority classifier learns its class from training only, with class zero in a tie. Logistic regression uses a training-only `StandardScaler` pipeline, $C = 1$, L2 regularization, `lbfgs`, balanced weights, tolerance 10^{-4} and at most 1,000 iterations. Threshold 0.5 was fixed; no hyperparameter/threshold search used test results. The original forest exactly reproduced its saved provenance, indices, scores, predictions and metrics. The baselines were added after inspecting the forest, so the comparison is exploratory; it does not compare CSI with RSSI/RTI or camera-assisted IsukaW.

Physical checks preserve hashes, distinct files, non-overlapping acquisition intervals and intact rounds. They do not establish independent persons or replicates. Rejected windows are not absence predictions. No intervals assume 165 independent observations. Editor actors remain known geometry, not inferred species, counts or positions.

3.5. Reproducibility

The project repository (https://github.com/Camimilo-11/Ai-Yachay-2026/tree/main/Project-safe-wireless) holds the simulator, the EM calibration code, the evaluation scripts, the run configurations with seeds and code hashes, the raw synthetic data, a requirements file, and the generated predictions, summaries, and figures. A verification script checks the package; one command (`ejecutar_integracion.py --quick --suite`) runs the nine short scenarios, and a second command with the base configuration runs the full 12-node, 100-cell setup. Each run writes to a new output folder, and the README documents the steps. The required software is Python 3.10 or later and a C++17 compiler (g++); the Python scripts are the reproduction workflow, so no notebook is needed. All V1 data are synthetic; the measured extension is separate and its raw radio identifiers remain local.

The original RTI/EM project structure is retained. The measured extension currently resides in the local `safe-wirreles` working copy; integration and synchronization with the linked academic repository remain pending. The legacy environment requires Python 3.10 or later and C++17; the Sionna extension uses its separate pinned Python 3.12 environment. Commands and result paths below are relative to the extension root. This avoids mixing incompatible environment assumptions.

The extension preserves firmware/data hashes, manifests, JSON models, window quality and fold predictions. Summaries are `results/esp32/presence-model-20261030.json` and `results/esp32/presence-baselines-20261030.json`. The original model summary identifies commit `544fe3f33ab97b7cecdf80da82d9f03d7de4063b`.

The public derived dataset `results/esp32/public-presence-features-20261030.npz` contains exactly 165 rows and twelve numeric features, labels, anonymous round codes and conditions, with a JSON metadata companion. The reproduction command is `scripts/reproduce_presence_from_features.py`; it re-fits the forest and both baselines

on the preserved folds and checks exact scores, predictions and metrics. This permits evaluation reproduction without raw network identifiers. It does not independently reproduce packet parsing, filtering, empty calibration, interpolation or feature extraction; those stages require the retained private raw files and manifests. Public metadata explicitly distinguishes redacted provenance from the private manifest. The original synthetic data and measured derived data remain separate.

```
# From the local safe-wirreles extension root:
$repro = @(
  "scripts/reproduce_presence_from_features.py"
  "--dataset"
  "results/esp32/public-presence-features-20261030.npz"
  "--metadata"
  "results/esp32/public-presence-features-20261030.json"
  "--output"
  "output/public-presence-reproduction.json"
)
.\.venv\Scripts\python.exe @repro
```

4. Preliminary Results

The quick base run used seed 7, six nodes, 15 links, 36 cells, and 30 test queries. EM converged after 294 iterations using 900 calibration residuals. It estimated $\hat{p} = 0.541$, $\hat{\sigma}_1 = 1.060$ dB, and $\hat{\sigma}_2 = 2.938$ dB. For 450 held-out residuals, the mixture's mean log density was -2.101; the single Gaussian's was -2.145. The difference is small and comes from one generated split. Table 3 reports the localization results.

Table 3. Localization results of the quick base run (seed 7, 30 test queries).

Method	Mean Localization Error (m)	RMSE (m)	Coverage
Tikhonov RTI	3.653	4.113	30/30
EM-derived Fisher RTI	3.530	4.015	30/30
Matched Tikhonov	3.530	4.015	30/30
Weighted K-NN	2.529	2.836	30/30
Constant room center	2.357	2.517	30/30

The center prediction has the lowest mean error in this pilot, while K-NN performs better than either RTI setting. The Fisher-based RTI and its equivalent Tikhonov control agree to numerical precision, as expected from the equations. In the stress suite, EM did not converge within 600 iterations when temporal correlation was set to $\rho = 0.8$. At 5 GHz with a low receiver margin, only 1.3% of test measurements remained valid, so the run was marked insufficient for calibration. These outcomes come from simulated conditions, not a measured radio channel.

The files under `results/integracion_verificada/` contain the reported values. The table uses `base_r00/summary.json` and can be regenerated with the command in the README. This is an executed simulation with one seed; it is not a physical experiment.

The larger V1 base configuration was also preserved: twelve nodes, 66 links, 100 cells and 120 test queries. Mean errors were 2.333 m for Tikhonov, 2.433 m for Fisher RTI and its matched control, 2.408 m for K-NN, and 2.483 m for the center. Geometry and sample size differ from the quick pilot and are not pooled. An older 500-query audit does not replace either canonical V1 result.

The artificial CSI ablation gave macro F1 0.9143 for signal and signal plus material descriptors. Brier changed from 0.1062 to 0.1056, while `other_moving` predicted as `human` increased from 0/60 to 20/60. This adverse result is preserved without claiming improvement. An initial physical empty/walking pair showed a 20.94% increase in median within-second RMS-amplitude coefficient of variation, with overlapping distributions: a descriptive contrast, not classification accuracy.

The physical model accepted 165 of 183 windows: twelve were rejected for gaps and six for boundary coverage. There were 84 empty and 81 person-present windows (42 still, 39 walking), from three rounds in one campaign; these are not 165 persons or independent experiments. Table 4 compares models on identical accepted windows.

Table 4. Exploratory physical CSI comparison using the same held-out-round predictions. Percentages except F1. PPV is undefined without positive predictions.

Model	PPV (%)	Recall (%)	F1	BA (%)	FPR (%)	Still recall (%)
Random forest	70.13	66.67	0.6835	69.64	27.38	61.90
Logistic regression	59.78	67.90	0.6358	61.93	44.05	66.67
Training majority	Undefined	0.00	0.0000	50.00	0.00	0.00

Forest counts were TN=61, FP=23, FN=27 and TP=54; logistic regression gave TN=47, FP=37, FN=26 and TP=55; majority gave TN=84, FP=0, FN=81 and TP=0. Walking sensitivity was 71.79% for the forest and 69.23% for logistic regression. Forest still-person sensitivity was 0%, 100% and 85.71% across rounds. The first fold's 100% PPV means only two correct alerts alongside 25 false negatives. The exported forest was later fitted on all three rounds; the evaluation forests fit two. A functional prediction check on training data was excluded from evaluation.

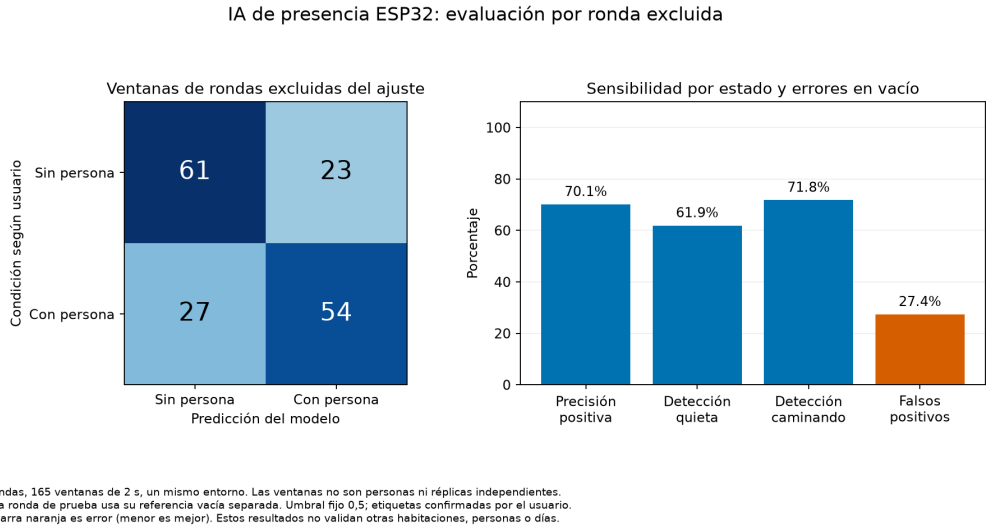


Figure 2. Physical CSI pilot. Left: reference rows and predicted columns. Right: forest PPV, still/walking sensitivity and empty-condition false positives (orange). There are 165 valid two-second windows from three rounds in one environment, each with its own separate empty reference. Windows are not independent persons. This results plot does not replace the student-designed system diagram.

5. Discussion and Expected Research Direction

The pilot shows that the mixture can be fitted and the RTI and fingerprint methods can run from generated CSV files. It also separates two different outcomes: the mixture

has a slightly better held-out residual score, but none of the signal-based methods beats the room-center prediction in mean location error. The Fisher-based RTI offers no change beyond its effective regularization; the matched Tikhonov control produces the same result.

The next stage should repeat each scenario with several seeds and report variation in both residual scores and location errors. It should also measure parameter recovery, since the simulator's true mixture parameters are known, and check predictive fit under quantization, drift, missing links, and correlation. A future extension could estimate link-specific responsibilities together with the attenuation field, following the broader direction of noise-adaptive RTI [4]. That would require a separate implementation and evaluation.

Within the V1 synthetic RTI layer, future presence tests could add independent cases with and without an attenuation target and choose a threshold using development data. That RTI extension remains unimplemented: its test queries all contain a target. The new physical CSI presence experiment is a separate observed extension, not evidence that RTI now performs presence detection. It would still represent an idealized target without debris or material properties. Evaluating people under rubble would require a separate, subsequent study with suitable physical measurements, sensing geometry, and validation.

The pilot has several limits. Its base case uses synthetic independent noise; fingerprint and test locations share one grid; and each query contains one stationary target. The V1 propagation model uses geometry rather than electromagnetic simulation, so reflections, diffraction, wall absorption, IEEE 802.11 MAC behavior and CSI are outside that layer. V2 separately adds static ray tracing and measured CSI, without changing the meaning of V1 data. Results from measured RSS and CSI studies provide context but do not predict this simulator's performance [1,4,8].

The applied comparison gives higher balanced accuracy and fewer empty-condition false positives for the forest than for the evaluated logistic model in this campaign, while logistic regression detects more still windows. High false positives and fold variation prevent a reliable-deployment claim. The comparison was added after the forest was inspected; it is not independent confirmation, significance testing or evidence of superiority over RSSI/RTI, another repository, new rooms or rescue sensing. Subject, geometry, traffic, gain and external motion were not independently controlled; a specific cause of errors cannot be established.

The defensible contribution is continuity of EM/RTI with expanded observability, physical acquisition and complete-round evaluation, including negative results. Simulated/real domain mismatch is a hypothesis, not a demonstrated cause. New days, rooms, recorded positions, still persons and non-person disturbances are needed, with models/thresholds frozen before test. There are no validated models for species, counting, real localization, poses, tissues or vital signs. The real binary model is not integrated into the live editor. Both research-parameter fields remain null. RuView is only a visual reference, with no borrowed accuracy claims.

The software record reports 393 passed tests and 33 passed subtests with Ruff and dependency checks; that does not establish detector reliability. A native intermittent LLVM/JIT failure remains unresolved despite later successful Sionna calculations. Position-likelihood RTI and repeated-seed validation remain planned work.

6. Conclusions

Project Safe Wireless preserves its EM-calibrated RSS/RTI study and controls, and extends the same paper with traceable propagation and a first physical CSI pilot. Better noise fit does not guarantee lower localization error. The new pilot demonstrates acquisition and grouped evaluation, with an exploratory matched-task comparison and unstable

performance. The contribution is an expanded reproducible experiment, not a validated detector or replacement for V1. Rescue under rubble remains a motivation requiring a separate physical study.

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Institutional Review Board Statement: The V1 synthetic study involved no human measurements. V2 includes student-confirmed presence conditions; the author must confirm applicable institutional review and consent requirements before dissemination. No institutional approval is asserted here.

Data Availability Statement: The original synthetic generator, C++17 EM, configurations, data and results remain in Project Safe Wireless: https://github.com/Camimilo-11/Ai-Yachay-2026/tree/main/Project-safe-wireless. The local measured extension includes a derived NPZ/JSON containing the same 165 feature rows, labels and anonymous group/condition codes, allowing exact refitting and evaluation without raw packets. This does not independently reproduce raw parsing, filtering, calibration, interpolation or extraction. Raw CSI with radio identifiers and original manifests remains local. Integrating and synchronizing this extension with the linked repository is pending; source and manuscript must correspond before submission.

Use of Artificial Intelligence: Generative AI was used to assist with language editing, preparation of the LaTeX document, generation of the web page used to simulate and visualize the scenarios, and correction of the scripts. The simulation data and preliminary numerical results were produced by running the scripts, not generated by AI. V2 also used AI assistance for code/document review and drafting.

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